GATE Electrical Engineering - Mock Test Set 3

Q.1 A single-phase voltage source $v_{s}=325~\sin(2\pi 50t)V$ delivers a current, $i=12~\sin(2\pi 50t)+9~\sin(2\pi 150t)$ A to a load. The load power factor, correct up to two decimal places, is:

(A) 1.00

(B) 0.80

(C) 0.65

(D) 0.57

Q.2 A 20 kVA, 1100 V/220 V, single-phase two-winding transformer is configured as a 1.32 kV/1.1 kV autotransformer. What will be the rating of the autotransformer?

(A) 60 KVA

(B) 75 KVA

(C) 90 kVA

(D) 100 kVA

Q.3 The Laplace transform of the step response of a system is given by $Y(s)=\frac{100}{s(s+100)}$. The rise time is defined as the time taken for the response to go from 0.1 to 0.9 of its final value. The settling time is defined as the time taken for the response to reach 0.98 of its final value. For this system, the rise time (T_{r}), settling time (T_{s}), and time constant (T_{c}) all expressed in seconds are

(A) $T_{r}=0.022$, $T_{s}=0.04$, $T_{c}=0.01$

(B) $T_{r}=0.22$, $T_{s}=0.404$, $T_{c}=0.01$

(C) $T_{r}=22$, $T_{s}=40.4$, $T_{c}=10.1$

(D) $T_{r}=2.2$, $T_{s}=4.04$, $T_{c}=1.01$

Q.4 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}}+7t\frac{dy}{dt}+8ty=10~\sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.5 Let $x_{c}(t)$ be any continuous - time periodic signal with period T. It is sampled uniformly with a sampling period T where $T_{s}\ne T$ resulting in the discrete sequence. $x[n]=x_{c}(nT_{s})$, where n is an integer. Which one of the following statements is correct about $x[n]$?

(A) $x[n]$ will always be periodic with period T/T_{x} for all values of T/T_{x}

(B) $x[n]$ will always be periodic with period 1 for all values of T/T_{s}

(C) $x[n]$ will never be periodic

(D) $x[n]$ will be periodic if and only if T/T_{s} is a rational number

Q.6 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.7 Consider a power system with N buses, of which P are generator buses and the remaining Q are load buses (where there is no generation). Assume that there are no reactive power-limit violations at the generator buses. What is the size of the Jacobian matrix in the Newton-Raphson load flow method?

(A) $2N\times2N$

(B) $(2N-1-P)\times(2N-1-P)$

(C) $(2N-Q)\times(2N-Q)$

(D) $(P+Q)\times(P+Q)$

Q.8 A positive point charge with velocity $\vec{v}=5\hat{x}$ enters a region having electric field $\vec{E}=4\hat{y}$ and magnetic field $\vec{B}=-6\hat{z}$. Which one of the following statements is correct for the force on the charge as it enters the region?

(A) The force will be along the magnetic field but perpendicular to the electric field

(B) The force will be along the electric field but perpendicular to the magnetic field

(C) The force will be perpendicular to both electric and magnetic field

(D) The magnetic field does not exert any force on the charge

Q.9 The Laplace transform of the step response of a system is given by $Y(s)=\frac{100}{s(s+100)}$. The rise time is defined as the time taken for the response to go from 0.1 to 0.9 of its final value. The settling time is defined as the time taken for the response to reach 0.98 of its final value. For this system, the rise time (T_{r}), settling time (T_{s}), and time constant (T_{c}) all expressed in seconds are

(A) $T_{r}=0.022$, $T_{s}=0.04$, $T_{c}=0.01$

(B) $T_{r}=0.22$, $T_{s}=0.404$, $T_{c}=0.01$

(C) $T_{r}=22$, $T_{s}=40.4$, $T_{c}=10.1$

(D) $T_{r}=2.2$, $T_{s}=4.04$, $T_{c}=1.01$

Q.10 Two nn x matrices A and B have a common eigenvalues 2, and the same corresponding nonzero eigen vector. Which of the following options is/are correct? (Note : I is the nn x identity matrix)

(A) Determinant (2) 0 AI -=

(B) Determinant (2) 0 BI -=

(C) Determinant (2) 0 A B I + - =

(D) Determinant (4) 0 A B I + - =

Q.11 A positive point charge with velocity $\vec{v}=5\hat{x}$ enters a region having electric field $\vec{E}=4\hat{y}$ and magnetic field $\vec{B}=-6\hat{z}$. Which one of the following statements is correct for the force on the charge as it enters the region?

(A) The force will be along the magnetic field but perpendicular to the electric field

(B) The force will be along the electric field but perpendicular to the magnetic field

(C) The force will be perpendicular to both electric and magnetic field

(D) The magnetic field does not exert any force on the charge

Q.12 Two nn x matrices A and B have a common eigenvalues 2, and the same corresponding nonzero eigen vector. Which of the following options is/are correct? (Note : I is the nn x identity matrix)

(A) Determinant (2) 0 AI -=

(B) Determinant (2) 0 BI -=

(C) Determinant (2) 0 A B I + - =

(D) Determinant (4) 0 A B I + - =

Q.13 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}}+7t\frac{dy}{dt}+8ty=10~sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.14 A uniform ring charge of radius R carries a total charge Q. Which one of the following options correctly quantifies the magnitude of the force on a point charge of strength q kept at the center of the ring? (e is the permittivity of the medium)

(A) $\frac{Qq}{4\pi\in R}$

(B) $\frac{Qq}{4\pi\epsilon R^{2}}$

(C) 0

(D) $\frac{q}{4\pi\epsilon r}\times\frac{Q}{2\pi R}$

Q.15 Which one of the following options is correct regarding the typical double-squirrel-cage structure used in induction motors ?

(A) At starting, a larger portion of the rotor current flows in the outer cage

(B) At starting, a larger portion of the rotor current flows in the inner cage

(C) At rated speed, most of the rotor current flows in the outer cage

(D) The purpose of the double-squirrel-cage structure is to lower the effective rotor resistance at starting

Q.16 The Laplace transform of the step response of a system is given by $Y(s)=\frac{100}{s(s+100)}$. The rise time is defined as the time taken for the response to go from 0.1 to 0.9 of its final value. The settling time is defined as the time taken for the response to reach 0.98 of its final value. For this system, the rise time (T_{r}), settling time (T_{s}), and time constant (T_{c}) all expressed in seconds are

(A) $T_{r}=0.022$, $T_{s}=0.04, T_{c}=0.01$

(B) $T_{r}=0.22, T_{s}=0.404, T_{c}=0.01$

(C) $T_{r}=22$, $T_{s}=40.4, T_{c}=10.1$

(D) $T_{r}=2.2, T_{s}=4.04$, $T_{c}=1.01$

Q.17 A 20 kVA, 1100 V/220 V, single-phase two-winding transformer is configured as a 1.32 kV/1.1 kV autotransformer. What will be the rating of the autotransformer?

(A) 60 KVA

(B) 75 KVA

(C) 90 kVA

(D) 100 kVA

Q.18 Let $x_{c}(t)$ be any continuous - time periodic signal with period T. It is sampled uniformly with a sampling period T where $T_{s}\ne T$ resulting in the discrete sequence. $x[n]=x_{c}(nT_{s})$, where n is an integer. Which one of the following statements is correct about $x[n]$?

(A) $x[n]$ will always be periodic with period T/T_{x} for all values of T/T_{x}

(B) $x[n]$ will always be periodic with period 1 for all values of T/T_{s}

(C) $x[n]$ will never be periodic

(D) $x[n]$ will be periodic if and only if T/T_{s} is a rational number

Q.19 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.20 Consider a power system with N buses, of which P are generator buses and the remaining Q are load buses (where there is no generation). Assume that there are no reactive power-limit violations at the generator buses. What is the size of the Jacobian matrix in the Newton-Raphson load flow method?

(A) $2N\times2N$

(B) $(2N-1-P)\times(2N-1-P)$

(C) $(2N-Q)\times(2N-Q)$

(D) $(P+Q)\times(P+Q)$

Q.21 A certain application requires power at a frequency of 16.67 Hz, while the available grid frequency is 50 Hz. A 3-phase synchronous motor connected to the 50 Hz, 3-phase grid, driving a synchronous generator, is used for this application. Which one of the following combinations is a suitable choice for the number of poles in the motor and generator, respectively?

(A) 2, 6

(B) 6, 2

(C) 4, 6

(D) 6, 4

Q.22 Which one of the following options is correct regarding the typical double-squirrel-cage structure used in induction motors ?

(A) At starting, a larger portion of the rotor current flows in the outer cage

(B) At starting, a larger portion of the rotor current flows in the inner cage

(C) At rated speed, most of the rotor current flows in the outer cage

(D) The purpose of the double-squirrel-cage structure is to lower the effective rotor resistance at starting

Q.23 A certain application requires power at a frequency of 16.67 Hz, while the available grid frequency is 50 Hz. A 3-phase synchronous motor connected to the 50 Hz, 3-phase grid, driving a synchronous generator, is used for this application. Which one of the following combinations is a suitable choice for the number of poles in the motor and generator, respectively?

(A) 2, 6

(B) 6, 2

(C) 4, 6

(D) 6, 4

Q.24 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.25 Let $x_{c}(t)$ be any continuous - time periodic signal with period T. It is sampled uniformly with a sampling period T where $T_{s}\ne T$ resulting in the discrete sequence. $x[n]=x_{c}(nT_{s})$, where n is an integer. Which one of the following statements is correct about $x[n]$?

(A) $x[n]$ will always be periodic with period T/T_{x} for all values of T/T_{x}

(B) $x[n]$ will always be periodic with period 1 for all values of T/T_{s}

(C) $x[n]$ will never be periodic

(D) $x[n]$ will be periodic if and only if T/T_{s} is a rational number

Q.26 Consider a power system with N buses, of which P are generator buses and the remaining Q are load buses (where there is no generation). Assume that there are no reactive power-limit violations at the generator buses. What is the size of the Jacobian matrix in the Newton-Raphson load flow method?

(A) $2N\times2N$

(B) $(2N-1-P)\times(2N-1-P)$

(C) $(2N-Q)\times(2N-Q)$

(D) $(P+Q)\times(P+Q)$

Q.27 Which one of the following options is correct regarding the typical double-squirrel-cage structure used in induction motors ?

(A) At starting, a larger portion of the rotor current flows in the outer cage

(B) At starting, a larger portion of the rotor current flows in the inner cage

(C) At rated speed, most of the rotor current flows in the outer cage

(D) The purpose of the double-squirrel-cage structure is to lower the effective rotor resistance at starting

Q.28 The causal signal with z-transform $z^{2}(z-a)^{-2}$ is $(u[n]$ is the unit step signal)

(A) $a^{2n}u[n]$

(B) $(n+1)a^{n}u[n]$

(C) $n^{-1}a^{n}u[n]$

(D) $n^{2}a^{n}u[n]$

Q.29 The causal signal with z-transform $z^{2}(z-a)^{-2}$ is $(u[n]$ is the unit step signal)

(A) $a^{2n}u[n]$

(B) $(n+1)a^{n}u[n]$

(C) $n^{-1}a^{n}u[n]$

(D) $n^{2}a^{n}u[n]$

Q.30 A certain application requires power at a frequency of 16.67 Hz, while the available grid frequency is 50 Hz. A three-phase synchronous motor connected to the 50 Hz, three-phase grid, driving a synchronous generator, is used for this application. Which one of the following combinations is a suitable choice for the number of poles in the motor and generator, respectively?

(A) 2, 6

(B) 6, 2

(C) 4, 6

(D) 6, 4

Q.31 A positive point charge with velocity $\vec{v}=5\hat{x}$ enters a region having electric field $\vec{E}=4\hat{y}$ and magnetic field $\vec{B}=-6\hat{z}$. Which one of the following statements is correct for the force on the charge as it enters the region?

(A) The force will be along the magnetic field but perpendicular to the electric field

(B) The force will be along the electric field but perpendicular to the magnetic field

(C) The force will be perpendicular to both electric and magnetic field

(D) The magnetic field does not exert any force on the charge

Q.32 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.33 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}}+7t\frac{dy}{dt}+8ty=10~sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.34 Which one of the following options is correct regarding the typical double-squirrel-cage structure used in induction motors ?

(A) At starting, a larger portion of the rotor current flows in the outer cage

(B) At starting, a larger portion of the rotor current flows in the inner cage

(C) At rated speed, most of the rotor current flows in the outer cage

(D) The purpose of the double-squirrel-cage structure is to lower the effective rotor resistance at starting

Q.35 Which one of the following options is correct regarding the typical double-squirrel-cage structure used in induction motors ?

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(C) At rated speed, most of the rotor current flows in the outer cage

(D) The purpose of the double-squirrel-cage structure is to lower the effective rotor resistance at starting

Q.36 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.37 A positive point charge with velocity $\vec{v}=5\hat{x}$ enters a region having electric field $\vec{E}=4\hat{y}$ and magnetic field $\vec{B}=-6\hat{z}$. Which one of the following statements is correct for the force on the charge as it enters the region?

(A) The force will be along the magnetic field but perpendicular to the electric field

(B) The force will be along the electric field but perpendicular to the magnetic field

(C) The force will be perpendicular to both electric and magnetic field

(D) The magnetic field does not exert any force on the charge

Q.38 A positive point charge with velocity $\vec{v}=5\hat{x}$ enters a region having electric field $\vec{E}=4\hat{y}$ and magnetic field $\vec{B}=-6\hat{z}$. Which one of the following statements is correct for the force on the charge as it enters the region?

(A) The force will be along the magnetic field but perpendicular to the electric field

(B) The force will be along the electric field but perpendicular to the magnetic field

(C) The force will be perpendicular to both electric and magnetic field

(D) The magnetic field does not exert any force on the charge

Q.39 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}} +7t\frac{dy}{dt}+8ty=10~sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.40 Two nn x matrices A and B have a common eigenvalues 2, and the same corresponding nonzero eigen vector. Which of the following options is/are correct? (Note : I is the nn x identity matrix)

(A) Determinant (2) 0 AI -=

(B) Determinant (2) 0 BI -=

(C) Determinant (2) 0 A B I + - =

(D) Determinant (4) 0 A B I + - =

Q.41 A 220V/12V single-phase transformer is designed for use in India and rated 150 VA at 60 Hz. Later, this unit is shipped to the USA where it is used as a 110V/6V transformer at 60 Hz. Which of the following statements is/are correct ?

(A) No-load current drawn would be smaller for operation in the USA compared to that in India

(B) For the same load current, the losses would be higher in the USA compared to that in India

(C) The peak magnetic flux density in the core would be higher while operating in the USA compared to that in India

(D) The eddy current losses in the core would be approximately 44% higher while operating in the USA compared to that in India

Q.42 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}}+7t\frac{dy}{dt}+8ty=10~sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.43 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.44 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}}+7t\frac{dy}{dt}+8ty=10~sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.45 Consider a power system with N buses, of which P are generator buses and the remaining Q are load buses (where there is no generation). Assume that there are no reactive power-limit violations at the generator buses. What is the size of the Jacobian matrix in the Newton-Raphson load flow method?

(A) $2N\times2N$

(B) $(2N-1-P)\times(2N-1-P)$

(C) $(2N-Q)\times(2N-Q)$

(D) $(P+Q)\times(P+Q)$

Q.46 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.47 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.48 A 25 kVA, 1100 V/220 V, single-phase two-winding transformer is configured as a 1.32 kV/1.1 kV autotransformer. What will be the rating of the autotransformer?

(A) 60 KVA

(B) 75 KVA

(C) 90 kVA

(D) 100 kVA

Q.49 Two nn x matrices A and B have a common eigenvalues 2, and the same corresponding nonzero eigen vector. Which of the following options is/are correct? (Note : I is the nn x identity matrix)

(A) Determinant (2) 0 AI -=

(B) Determinant (2) 0 BI -=

(C) Determinant (2) 0 A B I + - =

(D) Determinant (4) 0 A B I + - =

Q.50 A positive point charge with velocity $\vec{v}=5\hat{x}$ enters a region having electric field $\vec{E}=4\hat{y}$ and magnetic field $\vec{B}=-6\hat{z}$. Which one of the following statements is correct for the force on the charge as it enters the region?

(A) The force will be along the magnetic field but perpendicular to the electric field

(B) The force will be along the electric field but perpendicular to the magnetic field

(C) The force will be perpendicular to both electric and magnetic field

(D) The magnetic field does not exert any force on the charge

Q.51 A single-phase voltage source $v_{s}=325~sin(2\pi50t)V$ delivers a current, $i=12~sin(2\pi50t)+9~sin(2\pi150t)$ A to a load. The load power factor, correct up to two decimal places, is:

(A) 1.00

(B) 0.80

(C) 0.65

(D) 0.57

Q.52 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.53 The Laplace transform of the step response of a system is given by $Y(s)=\frac{100}{s(s+100)}$. The rise time is defined as the time taken for the response to go from 0.1 to 0.9 of its final value. The settling time is defined as the time taken for the response to reach 0.98 of its final value. For this system, the rise time (T_{r}), settling time (T_{s}), and time constant (T_{c}) all expressed in seconds are

(A) $T_{r}=0.022$, $T_{s}=0.04,$ $T_{c}=0.01$

(B) $T_{r}=0.22,$ $T_{s}=0.404,$ $T_{c}=0.01$

(C) $T_{r}=22$, $T_{s}=40.4,$ $T_{c}=10.1$

(D) $T_{r}=2.2,$ $T_{s}=4.04$, $T_{c}=1.01$

Q.54 For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

(A) $\frac{G}{1+GH}$

(B) $\frac{GH}{1+GH}$

(C) $\frac{1}{1+GH}$

(D) $\frac{1}{1+G}$

Q.55 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}}+7t\frac{dy}{dt}+8ty=10~sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.56 Which one of the following options is correct regarding the typical double-squirrel-cage structure used in induction motors ?

(A) At starting, a larger portion of the rotor current flows in the outer cage

(B) At starting, a larger portion of the rotor current flows in the inner cage

(C) At rated speed, most of the rotor current flows in the outer cage

(D) The purpose of the double-squirrel-cage structure is to lower the effective rotor resistance at starting

Q.57 Let $x_{c}(t)$ be any continuous - time periodic signal with period T. It is sampled uniformly with a sampling period T where $T_{s}\ne T$ resulting in the discrete sequence. $x[n]=x_{c}(nT_{s})$, where n is an integer. Which one of the following statements is correct about $x[n]$?

(A) $x[n]$ will always be periodic with period T/T_{x} for all values of T/T_{x}

(B) $x[n]$ will always be periodic with period 1 for all values of T/T_{s}

(C) $x[n]$ will never be periodic

(D) $x[n]$ will be periodic if and only if T/T_{s} is a rational number

Q.58 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}}+7t\frac{dy}{dt}+8ty=10~sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.59 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}}+7t\frac{dy}{dt}+8ty=10~sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.60 A uniform ring charge of radius R carries a total charge Q. Which one of the following options correctly quantifies the magnitude of the force on a point charge of strength q kept at the center of the ring? (e is the permittivity of the medium)

(A) $\frac{Qq}{4\pi\in R}$

(B) $\frac{Qq}{4\pi\epsilon R^{2}}$

(C) 0

(D) $\frac{q}{4\pi\epsilon r}\times\frac{Q}{2\pi R}$

Q.61 The Laplace transform of the step response of a system is given by $Y(s)=\frac{100}{s(s+100)}$. The rise time is defined as the time taken for the response to go from 0.1 to 0.9 of its final value. The settling time is defined as the time taken for the response to reach 0.98 of its final value. For this system, the rise time (T_{r}), settling time (T_{s}), and time constant (T_{c}) all expressed in seconds are

(A) $T_{r}=0.022$, $T_{s}=0.04$, $T_{c}=0.01$

(B) $T_{r}=0.22$, $T_{s}=0.404$, $T_{c}=0.01$

(C) $T_{r}=22$, $T_{s}=40.4$, $T_{c}=10.1$

(D) $T_{r}=2.2$, $T_{s}=4.04$, $T_{c}=1.01$

Q.62 A uniform ring charge of radius R carries a total charge Q. Which one of the following options correctly quantifies the magnitude of the force on a point charge of strength q kept at the center of the ring? (e is the permittivity of the medium)

(A) $\frac{Qq}{4\pi\in R}$

(B) $\frac{Qq}{4\pi\epsilon R^{2}}$

(C) 0

(D) $\frac{q}{4\pi\epsilon r}\times\frac{Q}{2\pi R}$

Q.63 The causal signal with z-transform $z^{2}(z-a)^{-2}$ is ($u[n]$ is the unit step signal)

(A) $a^{2n}u[n]$

(B) $(n+1)a^{n}u[n]$

(C) $n^{-1}a^{n}u[n]$

(D) $n^{2}a^{n}u[n]$

Q.64 A positive point charge with velocity $\vec{v}=5\hat{x}$ enters a region having electric field $\vec{E}=4\hat{y}$ and magnetic field $\vec{B}=-6\hat{z}$. Which one of the following statements is correct for the force on the charge as it enters the region?

(A) The force will be along the magnetic field but perpendicular to the electric field

(B) The force will be along the electric field but perpendicular to the magnetic field

(C) The force will be perpendicular to both electric and magnetic field

(D) The magnetic field does not exert any force on the charge

Q.65 A certain application requires power at a frequency of 16.67 Hz, while the available grid frequency is 50 Hz. A 3-phase synchronous motor connected to the 50 Hz, 3-phase grid, driving a synchronous generator, is used for this application. Which one of the following combinations is a suitable choice for the number of poles in the motor and generator, respectively?

(A) 2, 6

(B) 6, 2

(C) 4, 6

(D) 6, 4

Q.66 A uniform ring charge of radius R carries a total charge Q. Which one of the following options correctly quantifies the magnitude of the force on a point charge of strength q kept at the center of the ring? (e is the permittivity of the medium)

(A) $\frac{Qq}{4\pi\in R}$

(B) $\frac{Qq}{4\pi\epsilon R^{2}}$

(C) 0

(D) $\frac{q}{4\pi\epsilon r}\times\frac{Q}{2\pi R}$

Q.67 A 220V/12V single-phase transformer is designed for use in India and rated 200 VA at 50 Hz. Later, this unit is shipped to the USA where it is used as a 110V/6V transformer at 60 Hz. Which of the following statements is/are correct ?

(A) No-load current drawn would be smaller for operation in the USA compared to that in India

(B) For the same load current, the losses would be higher in the USA compared to that in India

(C) The peak magnetic flux density in the core would be higher while operating in the USA compared to that in India

(D) The eddy current losses in the core would be approximately 44% higher while operating in the USA compared to that in India

Q.68 A 15 kVA, 1100 V/220 V, single-phase two-winding transformer is configured as a 1.32 kV/1.1 kV autotransformer. What will be the rating of the autotransformer?

(A) 60 KVA

(B) 75 KVA

(C) 90 kVA

(D) 100 kVA

Q.69 Consider the following differential equation : $t^{2}\frac{d^{2}y}{dt^{2}}+7t\frac{dy}{dt}+8ty=10~sin(t)$. Which one of the following options is correct?

(A) It is a linear differential equation

(B) It is a nonlinear differential equation

(C) It is a time-invariant differential equation

(D) It is a second-order partial differential equation

Q.70 A 25 kVA, 1100 V/220 V, single-phase two-winding transformer is configured as a 1.32 kV/1.1 kV autotransformer. What will be the rating of the autotransformer?

(A) 60 KVA

(B) 75 KVA

(C) 90 kVA

(D) 100 kVA